



Assignment: 01

Assignment Title: Git, Gitea, GitHub, Git LFS, and GitHub Pages

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Section: B

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Task 01: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

```
tuba-zainab@ubuntu:~/Gitea$ docker compose ps
NAME                IMAGE              COMMAND                  SERVICE    CREATED          STATUS          PORTS
gitea                gitea/gitea:latest "/usr/bin/entrypoint..." gitea      21 minutes ago  Up 21 minutes  22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db             postgres:17-alpine "docker-entrypoint.s..." db         21 minutes ago  Up 21 minutes  5432/tcp
tuba-zainab@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
tuba-zainab@ubuntu:~/Gitea$ _
```

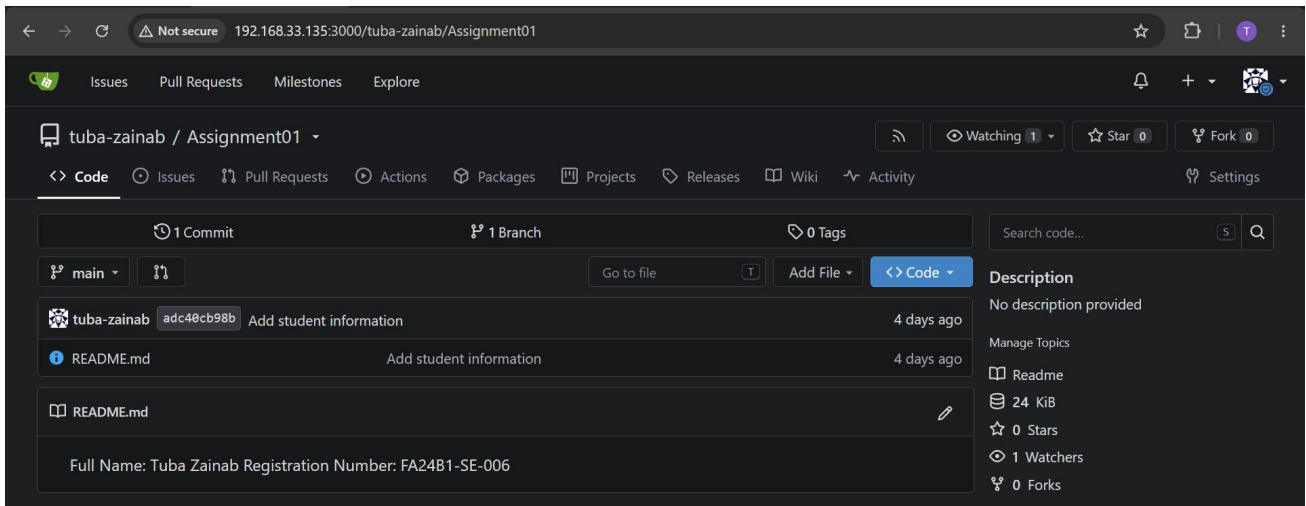
Docker and Docker Compose are installed on the Ubuntu server, and the provided Gitea configuration is used to start the service. The server and containers are checked to ensure Gitea is running correctly.

Step 1.2: Create and Push the Initial Repository

```
tuba-zainab@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': tuba-zainab
Password for 'https://tuba-zainab@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 283 bytes | 283.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tuba-zainab/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
tuba-zainab@ubuntu:~/Assignment01$ _
```

A local Git repository named Assignment01 is created with a README containing the student's information. The repository is connected to Gitea, and its initial commit is uploaded using a personal access token.

Step 1.3: Verify the Gitea Repository



The repository is opened in Gitea to check its name, public visibility, and README content. This confirms that the repository and student information are available on the Gitea server.

Result:

Gitea is installed and running, and the local repository is pushed to Gitea using HTTP over the server's local loopback interface.

Task 02: Push the Same Repository

Step 2.1: Create the GitHub Repository and Verify Both Remotes

```
tuba-zainab@ubuntu:~/Assignment01$ git remote -v
gitea    http://tuba-zainab@127.0.0.1:3000/tuba-zainab/Assignment01.git (fetch)
gitea    http://tuba-zainab@127.0.0.1:3000/tuba-zainab/Assignment01.git (push)
github   https://github.com/tuba-zainab/Assignment01.git (fetch)
github   https://github.com/tuba-zainab/Assignment01.git (push)
tuba-zainab@ubuntu:~/Assignment01$
```

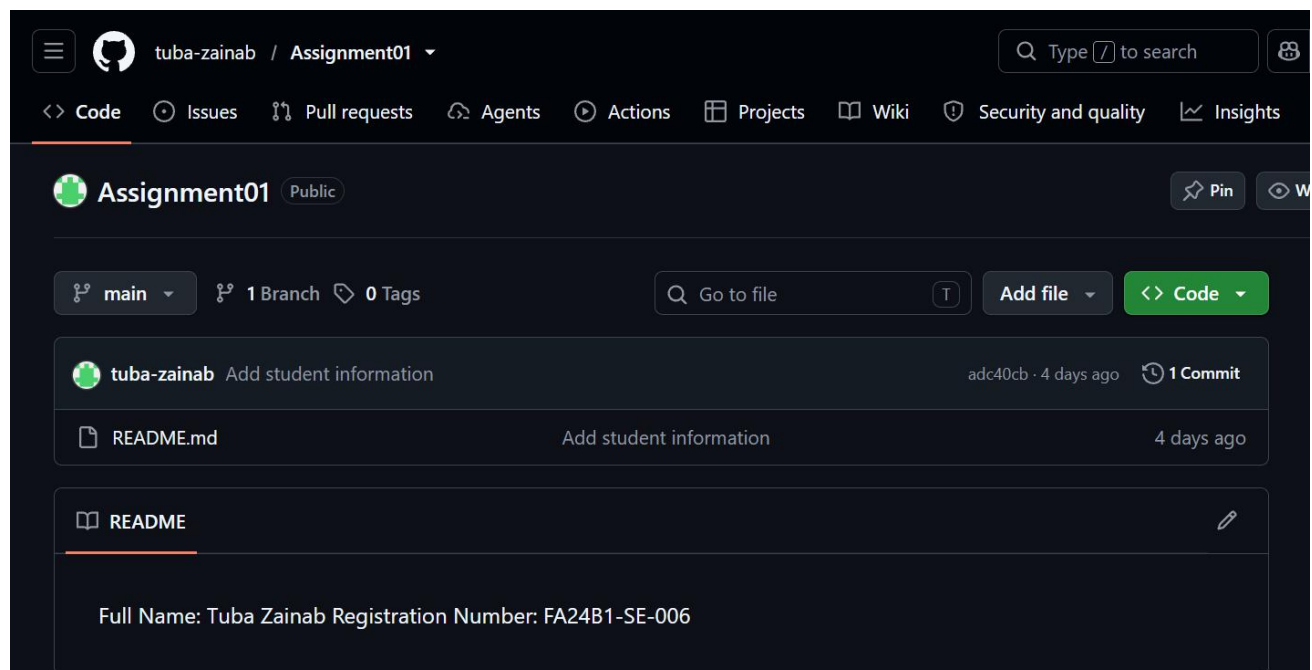
An empty public repository named Assignment01 is created on GitHub. GitHub is added as a second remote, and the remote list is checked to confirm that both Gitea and GitHub are connected.

Step 2.2: Push to GitHub

```
tuba-zainab@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': tuba-zainab
Password for 'https://tuba-zainab@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 283 bytes | 283.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tuba-zainab/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
tuba-zainab@ubuntu:~/Assignment01$
```

The main branch of the existing local repository is uploaded to GitHub. The terminal output is checked to confirm that the push completes successfully.

Step 2.3: Verify the GitHub Repository



The GitHub repository is opened in a browser to check its owner, name, public visibility, and README content. This verifies that the same project is available on GitHub.

Result:

The same local repository is connected to both Gitea and GitHub and is available publicly on GitHub.

Task 03: Track Three Large files with Git LFS

Step 3.1: Configure Git LFS

```
tuba-zainab@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
tuba-zainab@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
tuba-zainab@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
tuba-zainab@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
tuba-zainab@ubuntu:~/Assignment01$
```

Git LFS is initialized on the Ubuntu server, and its installed version is checked. Tracking is configured for .bin files so that Git LFS can manage files of this type.

Step 3.2: Create and Verify Three LFS Files

```
tuba-zainab@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
tuba-zainab@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
tuba-zainab@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
tuba-zainab@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
tuba-zainab@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
tuba-zainab@ubuntu:~/Assignment01$
```

Three files larger than 100 MiB are created and added to the repository along with the tracking configuration. Git LFS commands are used to verify that all three files are tracked and display their sizes.

Step 3.3: Commit and Push the LFS Files

```
tuba-zainab@ubuntu:~/Assignment01$ git commit -m "Add three files using Git LFS"
[main 3d8b6e8] Add three files using Git LFS
4 files changed, 10 insertions(+)
create mode 100644 .gitattributes
create mode 100644 large-file-1.bin
create mode 100644 large-file-2.bin
create mode 100644 large-file-3.bin
tuba-zainab@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': tuba-zainab
Password for 'https://tuba-zainab@github.com':
Username for 'https://github.com': tuba-zainab
Password for 'https://tuba-zainab@github.com':
Uploading LFS objects: 100% (3/3), 321 MB | 784 KB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 797 bytes | 265.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tuba-zainab/Assignment01.git
   adc40cb..3d8b6e8  main -> main
tuba-zainab@ubuntu:~/Assignment01$
```

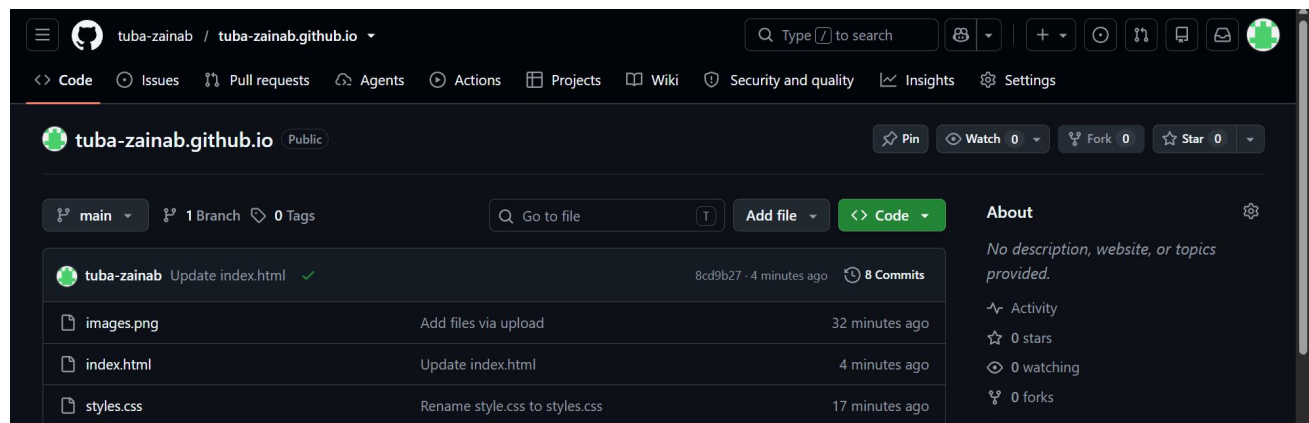
The tracking configuration and three large files are committed to Git. The changes are pushed to GitHub, and the output is checked for successful large-file uploads.

Result:

Three files larger than 100 MiB are tracked with Git LFS and pushed to the GitHub Assignment01 repository.

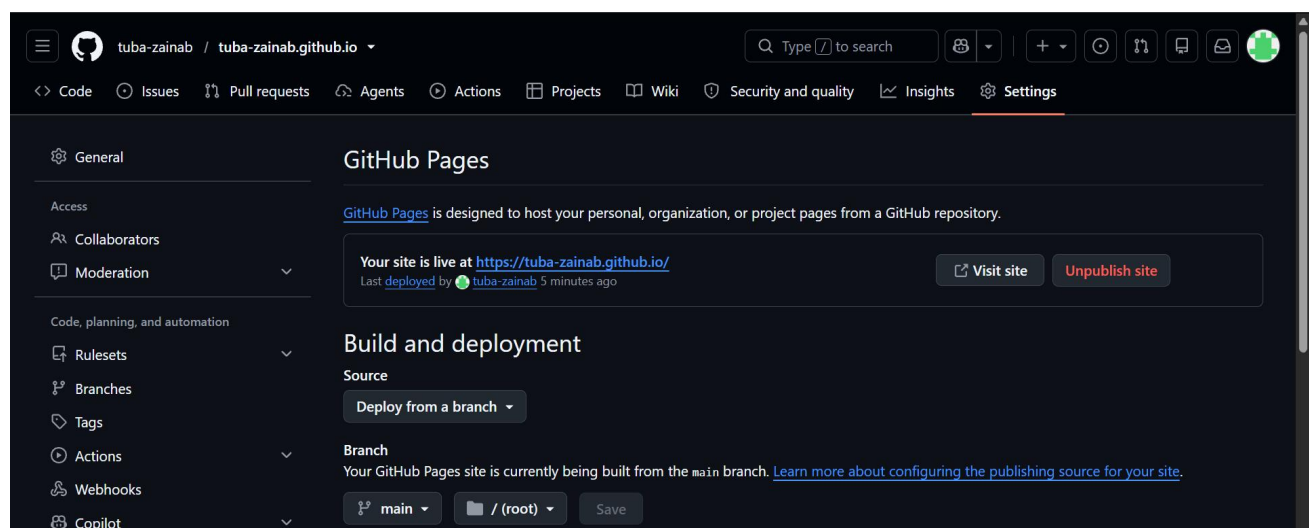
Task 04: Create a Portfolio or CV with GitHub Pages

Step 4.1: Create and Push the Portfolio Repository



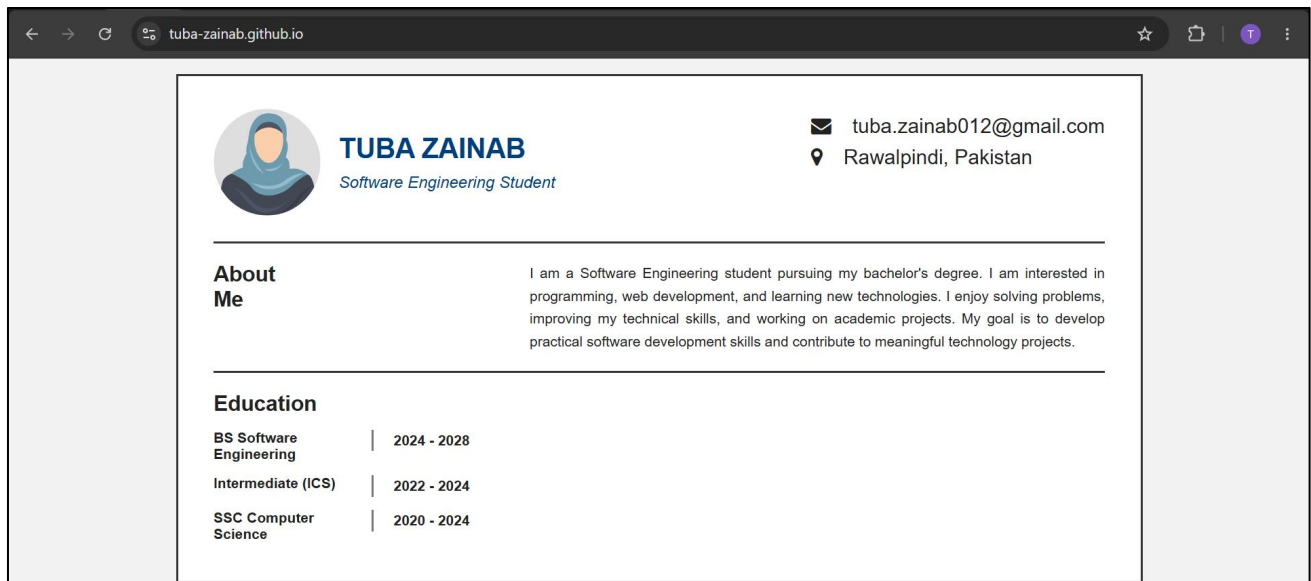
A public repository named username.github.io is created for the portfolio website. The HTML and CSS files are added with the required sections, then committed and pushed to the main branch.

Step 4.2: Publish with GitHub Pages



GitHub Pages is configured through the repository settings to deploy from the main branch and root folder. The deployment status and published website URL are checked.

Step 4.3: Verify the Live Portfolio



The published portfolio URL is opened in a browser to confirm that the website loads correctly. The portfolio content and website address are checked as evidence of successful publication.

Result:

The portfolio or CV is stored in a public GitHub repository and published as a live GitHub Pages site.